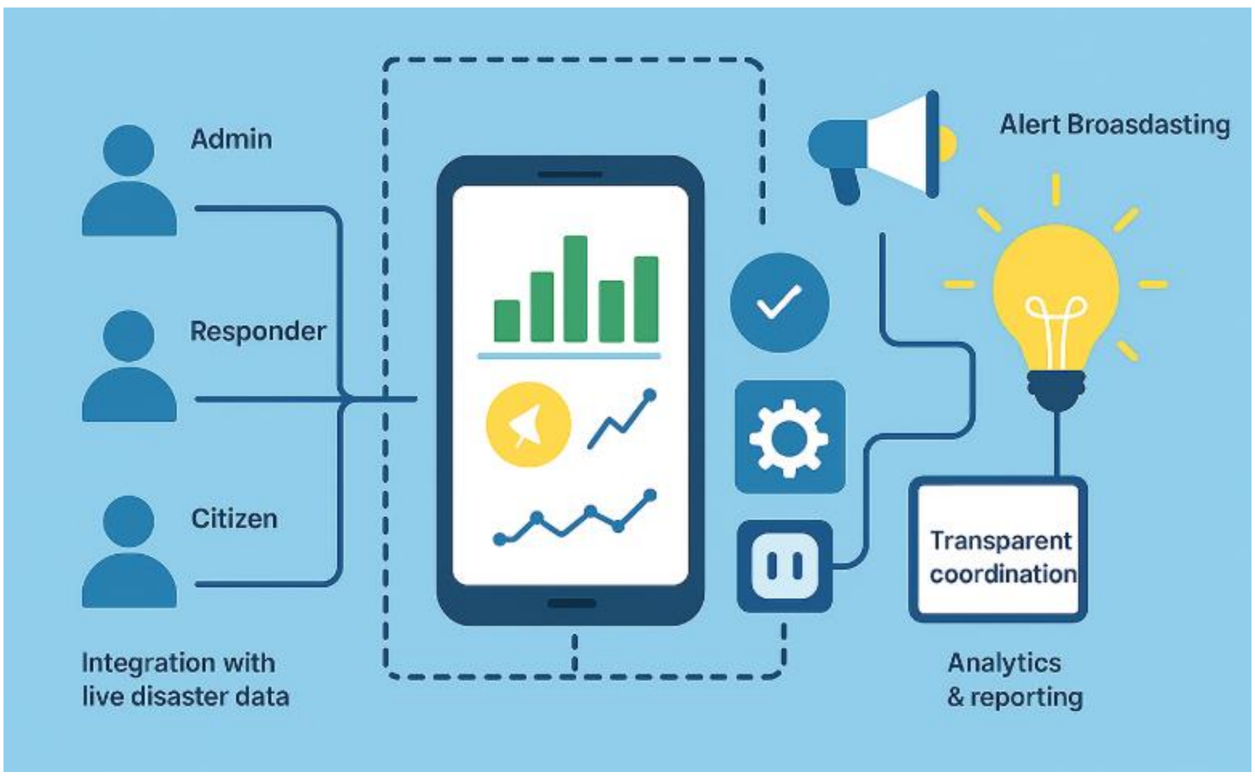
Disaster Management & Alert System

1.Project Statement

In disaster-prone regions, early warnings and efficient rescue coordination can save countless lives. However, the lack of real-time data sharing and communication between authorities, responders, and citizens often delays emergency response.

This project provides a centralized backend system to monitor disasters, send alerts, and manage rescue tasks efficiently.

It integrates live disaster data APIs, provides real-time notifications, and ensures transparent coordination among all roles.



2.Outcomes

- User Roles: **Admin, Responder, Citizen**
- Real-time alert broadcasting and acknowledgment system

- Live disaster data integration (e.g., floods, earthquakes, storms)
- Rescue operation management and task assignment
- Geo-mapped dashboard for tracking disaster zones
- Analytical reports on previous disasters and response times

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3.Modules to be Implemented

Authentication & Role Management

- JWT-based authentication for secure login and access control.
- Role-based permissions for Admin, Responder, and Citizen.
- Profile setup with personal details, contact info, and location.

Disaster Monitoring

- Integration with external APIs (e.g., OpenWeather, NDMA feeds) for real-time disaster data.
- Fetch and update current alerts by region.
- Categorize disasters (flood, earthquake, fire, cyclone, etc.).

Alert Management

- Admin can create, edit, or broadcast alerts.
- Citizens receive live notifications through email or in-app alerts.
- Responders acknowledge receipt and report progress.

Rescue Task Assignment

- Admin assigns responders to affected zones.

- Responders update task progress (Pending, Ongoing, Completed).
- Citizens can request rescue via form submission.

Analytics & Reporting

- Generate reports for disaster types, affected areas, and response times.
- Visual dashboards showing alert frequency and responder activity.
- Comparative data to evaluate preparedness and performance.

4.Week-Wise Milestones

Milestone 1: Weeks 1–2

Authentication & Role Setup

Implement JWT authentication:

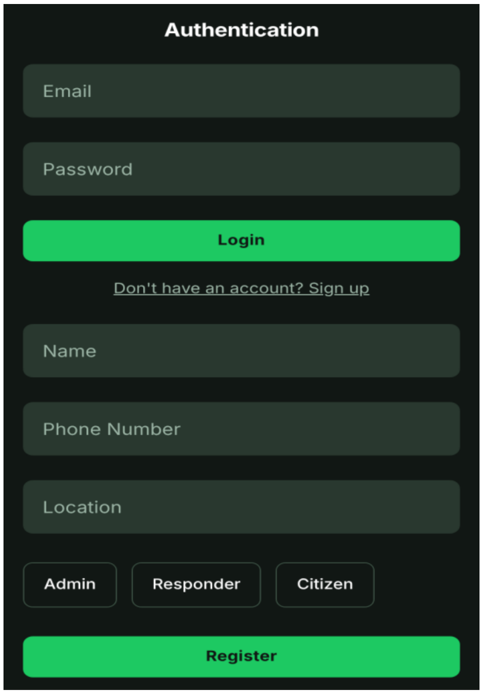
- Secure login and registration system using JSON Web Tokens (JWT).
- Ensure protected endpoints for role-based access and validation.

Role-based access (Admin, Responder, Citizen):

- Admin controls system configuration and task allocation.
- Responders manage assigned rescue operations.
- Citizens can view alerts and request help.

Profile setup with details and location:

- Capture user name, phone number, and region.
- Store data in the database for disaster-based notification targeting.



Milestone 2: Weeks 3–4

Disaster Monitoring
Integration with external APIs:

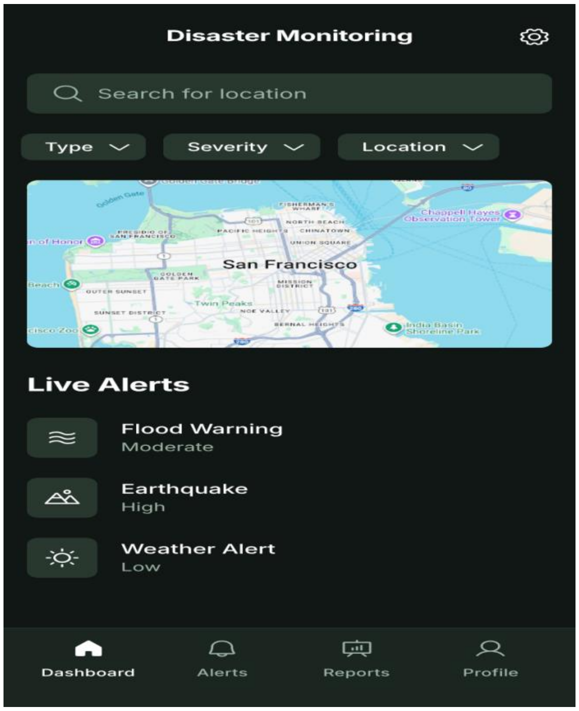
- Connect to live disaster alert APIs (e.g., weather, flood, earthquake).
- Parse and store real-time event data in the system.

Real-time alert dashboard:

- Display ongoing and past disasters on a dynamic dashboard.
- Include filtering by disaster type, severity, and location.

Disaster categorization:

- Automatically tag incoming data as “Flood”, “Cyclone”, etc.
- Allow Admins to verify and edit alert details before broadcast.



Milestone 3: Week 5

Alert Broadcasting & Management
Broadcast alerts:

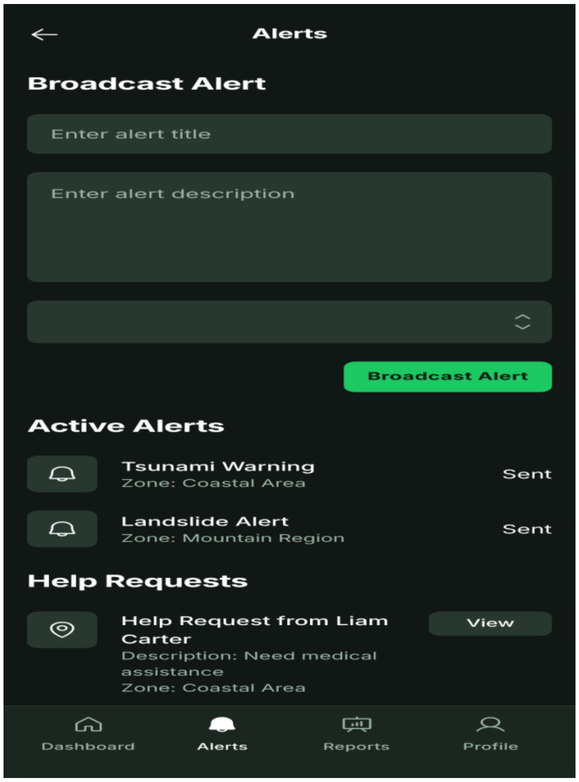
- Admin can push alerts to citizens within affected zones.
- Respective citizens receive instant notifications.

Acknowledge system:

- Responders must confirm receipt and readiness to act.
- Log acknowledgment times to track response efficiency.

Citizen reporting:

- Citizens can submit emergency help requests with location and description.
- The backend routes the request to the nearest available responder.



Milestone 4: Weeks 6–7

Rescue Operations Module
Task assignment & management:

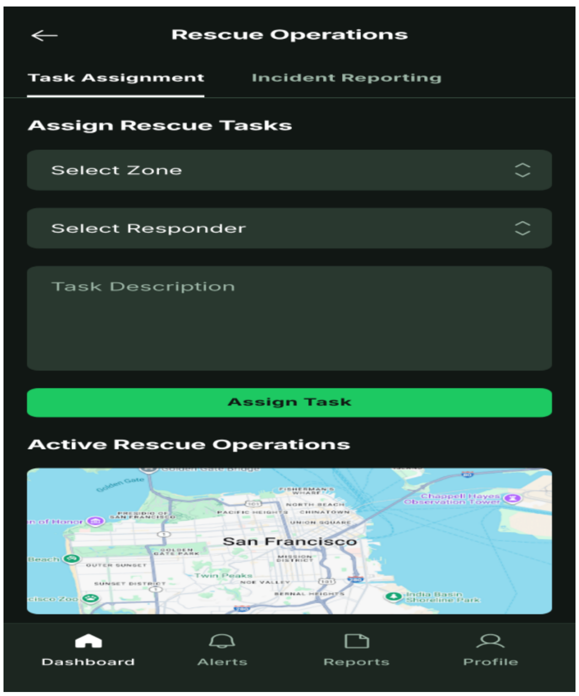
- Admin assigns rescue tasks to responders in specific zones.
- Responders can update task progress through their dashboard.

Geo-location integration:

- Store location coordinates of disaster zones and rescue sites.
- Map-based visualization for active rescue operations.

Incident reporting:

- Responders submit status reports and upload images (if supported).
- All reports are timestamped for audit and review.



Milestone 5: Week 8

Analytics & Reporting
Generate analytical dashboards:

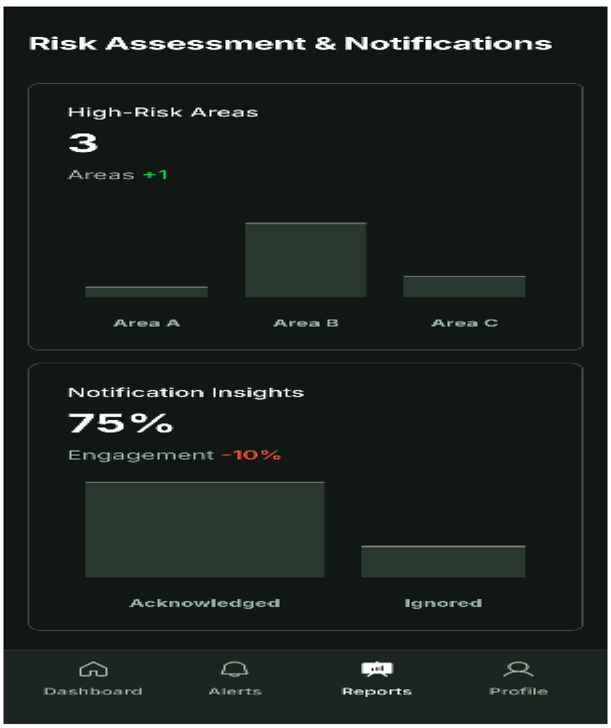
- Show the number of disasters handled per month/year.
- Compare response times and resolution efficiency by region.

Performance metrics:

- Evaluate responder activity and completion rates.
- Identify high-risk areas using historical data.

Notification insights:

- Track number of alerts broadcasted and user engagement levels.
- Summarize alerts acknowledged vs. ignored for improvement planning.



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5. Evaluation Criteria

- **Week 1:** Project setup completed with database schema and API skeleton initialized
- **Week 2:** Authentication & role-based access management fully functional
- **Week 3:** Disaster monitoring module integrated with external APIs (data successfully fetched

and stored)
- **Week 4:** Real-time disaster alert fetching and dashboard listing completed
- **Week 5:** Alert broadcasting and responder acknowledgment system tested successfully
- **Week 6:** Citizen reporting and request routing to responders implemented
- **Week 7:** Rescue task assignment and progress tracking fully operational
- **Week 8:** Analytics & reporting dashboard finalized and evaluated for accuracy

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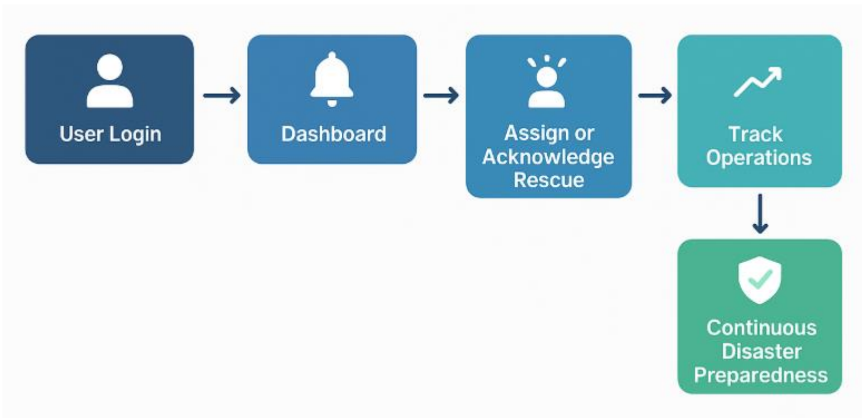
6. Workflow diagram:



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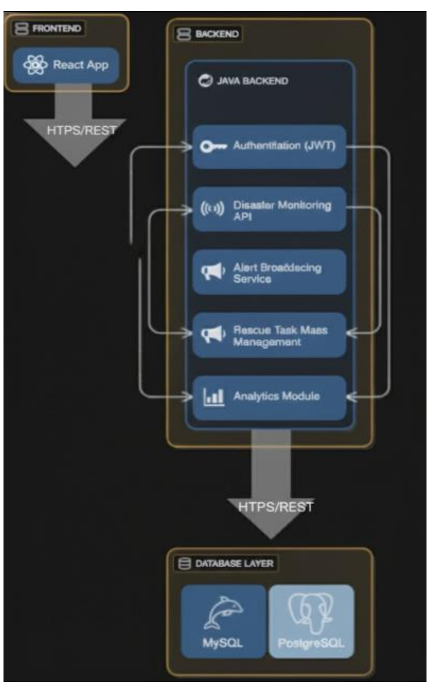
6. Architecture diagram:



8.Schema diagram: